

UNION COUNTY AP, OH, USA

WMO: 720651

Lat: 40.225N

Lon: 83.352W

Elev: 1021

StdP: 14.16

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00232

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	5.7	-7.6	3.8	0.2	-1.2	5.3	7.7	27.5	28.6	24.9	29.2	6.3	270	0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	89.9	72.5	87.7	71.4	85.5	70.5	75.8	84.5	74.5	82.9	73.2	81.4	9.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	127.8	80.3	72.0	123.1	79.3	70.5	116.7	77.8	40.0	84.5	38.7	83.1	37.4	81.5	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.3	19.4	17.2	DB	-8.3	93.2	8.3	3.1	-14.2	95.4	-19.1	97.2	-23.7	99.0	-29.7	101.2	
			WB	-8.8	78.3	8.2	1.7	-14.7	79.5	-19.5	80.5	-24.1	81.5	-30.1	82.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	26.0	29.7	39.2	51.5	63.7	70.7	73.5	70.6	65.2	53.6	41.1	33.2
	DBStd	18.81	12.19	12.46	10.82	8.83	8.85	5.67	5.18	4.88	7.55	9.07	9.22	9.80
	HDD50	2665	745	577	366	86	8	0	0	0	1	64	292	526
	HDD65	5741	1208	990	799	411	135	18	4	11	91	371	717	986
	CDD50	3252	1	7	32	130	431	620	730	639	457	175	25	5
	CDD65	856	0	0	0	6	93	188	269	186	97	17	0	0
	CDH74	7403	0	0	7	90	842	1539	2313	1502	929	177	4	0
	CDH80	2284	0	0	1	8	229	485	821	382	317	41	0	0
Wind	WSAvg	7.1	9.0	8.4	8.4	8.6	7.1	6.5	5.0	4.5	4.8	7.1	7.7	7.9
Precipitation	PrecAvg	40.2	3.0	2.2	2.9	3.7	4.6	4.5	4.4	3.5	3.0	2.6	2.8	3.0
	PrecMax	50.7	9.4	4.4	6.3	6.9	8.0	8.5	9.4	6.7	6.1	5.4	5.2	8.2
	PrecMin	27.2	1.2	0.8	0.9	1.1	2.0	1.7	1.8	0.9	0.7	0.7	0.8	0.9
	PrecStd	5.7	1.8	0.9	1.3	1.4	1.9	1.9	1.8	1.7	1.6	1.3	1.2	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	66.3	74.5	81.2	88.5	91.1	93.1	89.5	91.3	85.6	70.8	61.8
		MCWB	55.3	57.3	61.2	61.4	69.6	71.6	74.7	72.5	71.1	69.0	59.7	58.0
	2%	DB	54.4	58.7	67.4	76.7	85.6	88.3	90.1	86.6	87.7	78.8	65.1	56.8
		MCWB	51.3	54.2	57.7	60.4	67.6	70.8	74.2	72.2	70.8	64.3	55.4	53.4
	5%	DB	48.0	53.8	61.4	72.0	82.2	85.6	87.6	84.1	83.8	74.2	60.6	53.0
		MCWB	46.0	50.5	53.3	58.0	66.7	69.2	72.5	71.4	69.4	62.1	52.0	50.6
	10%	DB	43.0	46.5	56.8	67.2	79.0	82.3	84.7	81.8	79.5	69.4	55.9	47.9
		MCWB	40.1	42.5	51.1	56.6	65.1	68.2	71.6	70.0	66.2	60.7	49.7	45.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.5	59.6	62.2	65.6	72.3	75.5	78.2	77.2	74.9	70.9	62.1	58.9
		MCDB	59.1	65.2	72.8	77.0	82.8	85.5	88.1	84.3	85.1	80.8	69.2	61.4
	2%	WB	51.2	54.9	59.4	62.3	70.5	74.2	76.6	75.2	73.1	67.8	57.8	53.9
		MCDB	53.5	56.9	66.1	71.1	81.4	83.7	85.8	82.7	83.9	74.5	62.3	56.1
	5%	WB	46.3	49.7	55.4	60.3	68.8	72.3	75.0	73.8	71.2	65.0	53.6	50.5
		MCDB	47.7	54.6	59.2	68.4	79.0	81.2	83.3	80.6	80.2	70.9	58.7	52.9
	10%	WB	40.2	42.9	51.7	58.1	67.1	70.7	73.5	72.1	69.3	61.4	50.0	45.0
		MCDB	42.6	46.3	56.1	65.6	75.9	78.9	81.7	78.7	75.8	69.2	55.5	47.4
Mean Daily Temperature Range	5% DB	MDBR	15.1	16.1	17.8	21.8	20.5	19.4	19.2	20.9	22.6	21.2	18.2	13.3
		MCDBR	18.8	22.2	25.1	27.8	24.2	23.5	21.9	23.1	26.8	27.6	25.6	18.6
		MCWBR	16.9	17.9	15.7	14.9	9.9	9.6	8.4	10.6	11.5	14.1	16.6	16.6
	5% WB	MCDBR	18.8	21.4	21.0	22.8	20.2	19.4	19.1	18.1	22.0	21.7	20.8	17.7
		MCWBR	17.7	18.2	15.0	15.0	9.3	9.7	9.1	9.4	10.3	12.2	15.7	16.6
Clear-Sky Solar Irradiance	taub	0.307	0.326	0.355	0.395	0.435	0.448	0.454	0.449	0.391	0.359	0.326	0.301	
	taud	2.410	2.341	2.333	2.244	2.173	2.181	2.176	2.173	2.316	2.392	2.446	2.454	
	Ebn at Noon	271	281	284	279	269	265	261	259	268	266	262	265	
	Edh at Noon	26	32	36	43	47	46	46	45	36	30	25	23	
All-Sky Solar Radiation	RadAvg	516	800	1108	1495	1744	1917	1886	1723	1400	962	643	437	
	RadStd	48	96	102	128	144	138	110	80	121	84	83	44	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+122

Nomenclature: See separate page